

SAJIB RAFI HOSSAIN

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Education

Queens College, City University of New York

Bachelor's in Computer Science, Minor in Mathematics

Expected May 2027

Queens, New York

Technical Skills

Languages: JavaScript · Python · TypeScript · Java · C++ · HTML · CSS

Technologies/Frameworks: React.js · Next.js · Tailwind CSS · Firebase · Node.js · Express.js · Drizzle ORM · PostgreSQL · SQLite · DynamoDB · AWS · Supabase · Vitest · Git/Github · Bash Command Line

Experience

Software Engineer Fellow

July 2025 - Present

CUNY Tech Prep

Remote

- Selected for a competitive, year-long fellowship for top CUNY computer science students, focusing on full-stack web development, cloud technologies, and professional software engineering practices
- Training includes client- and server-side development, serverless & microservice APIs, ORMs, relational/NoSQL databases, TDD/BDD, CI/CD, security, Agile teamwork, and responsible AI agent integration
- Building full-stack applications with React/Next.js (TypeScript), Tailwind CSS, Node.js/Express, PostgreSQL, DynamoDB, Supabase, and AWS, supported by CI/CD pipelines with GitHub Actions
- Implementing engineering best practices, including minimum **70%** test coverage, Lambda API development, infrastructure testing with LocalStack, and robust documentation (ADRs, READMEs, API docs)

Software Engineer Fellow

July 2024 - September 2024

Headstarter AI

Remote

- Engineered and deployed **5** AI-powered web applications, targeting scalability and user growth
- Led a team in developing full-stack web applications using Next.js, React, Firebase, and other technologies, reducing development time by **30%** through implementing modular components and CI/CD workflows
- Optimized workflows to improve delivery speed and integrate OpenAI, Pinecone, Stripe, and Clerk seamlessly

Web Developer Fellow

September 2023 – April 2024

CodePath

Remote

- Expanded proficiency in HTML, CSS, JavaScript, and React.js through intensive project-based learning, including integrating external APIs and parsing JSON responses
- Collaborated in team-based coding sessions, improving project delivery through code reviews and pair programming

Projects

GreenBites | TypeScript, React, Next.js, Supabase

December 2025

- Developed a full-stack nutrition and sustainability tracker that lets users search foods via **Calorie Ninjas API**, adjust serving sizes, and track daily macros, calories, and carbon footprint with interactive dashboards.
- Designed **PostgreSQL** database schema on Supabase with **Drizzle ORM**, protected routes with Clerk authentication, resolved production deployment issues, and fixed persistence issues so tracking remains accurate across sessions.

CUNY GradeLens | JavaScript, React, Node.js, Express.js, OpenAI API, SQLite, Tailwind CSS

August 2025

- Built an AI-powered platform during the CTP Hackathon as part of a team of 3 that analyzes grade distributions with GPT-4, enabling **natural language queries** to deliver insights on professors and course difficulty.
- Developed a chatbot with conversation history, fuzzy name matching algorithms, and data visualizations (**Chart.js**), robust **CSV data ingestion**, regex-based entity extraction, and multi-level caching to optimize API costs by **60%**.

AI Powered Expense Tracker | JavaScript, React, Next.js, OpenAI API, Firebase, Clerk

September 2024

- Built an AI-powered expense tracker that allows users to split costs seamlessly using natural language input, with **OpenAI** extracting key details like amount, participants, and balances for automated calculations.
- Integrated Firebase for real-time data storage and **Clerk authentication**, ensuring secure access, persistent user data across sessions, and a seamless user experience with automated expense tracking and retrieval.

Leadership / Community Impact

STEM Instructor

July 2025 - August 2025

New York Edge

- Led environmental science and STEM lessons (sustainability, renewable energy, ecosystems, etc.) to **150+** grade 3–5 students; adapted instruction to diverse learners and fostered engagement through interactive, hands-on activities.